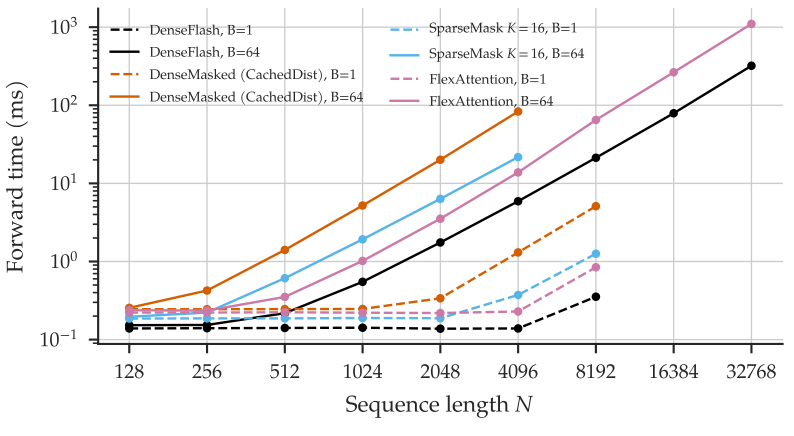
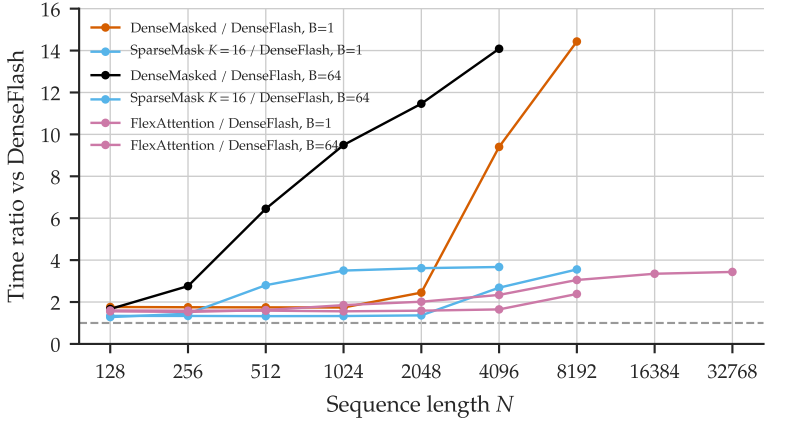

(a) Forward time: B=1 vs B=64
Forward time (ms)
Sequence length N
DenseFlash, B=1
DenseFlash, B=64
DenseMasked (CachedDist), B=1
DenseMasked (CachedDist), B=64
SparseMask K = 16, B=1
SparseMask K = 16, B=64
FlexAttention, B=1
FlexAttention, B=64
128
256
512
1024
2048
4096
8192
16384
32768


(b) Ratio vs unmasked dense
Time ratio vs DenseFlash
Sequence length N
DenseMasked / DenseFlash, B=1
SparseMask K = 16 / DenseFlash, B=1
DenseMasked / DenseFlash, B=64
SparseMask K = 16 / DenseFlash, B=64
FlexAttention / DenseFlash, B=1
FlexAttention / DenseFlash, B=64
128
256
512
1024
2048
4096
8192
16384
32768